SafeRide — Product Requirements Document (PRD)

Smart Helmet Strap Check & Crash SOS Add-On | Website PRD and VS Code Build Prompt

1. Product overview

Product: SafeRide — a low-cost smart helmet add-on that checks whether the helmet strap is fastened and sends an SOS alert with GPS location after a detected severe crash.

Website purpose: Explain the rider-safety problem, present the SafeRide solution, demonstrate the prototype, build credibility, and collect pilot-test interest.

Primary users: Daily commuters, students, delivery riders, long-distance riders, parents, colleges, and potential partners.

Target price: Rs 1,000–Rs 1,500 as an add-on module.

2. Problem and opportunity

Two-wheeler riders account for a large share of India’s road deaths. In 2024, MoRTH recorded 1,77,175 road fatalities, while NCRB recorded 84,599 two-wheeler deaths, or 48.3% of road-transport deaths. Helmet non-use was associated with 54,122 deaths in 2024.

A conventional helmet protects the head during a crash, but it cannot verify correct use or automatically request help afterward. SafeRide addresses both gaps through strap checking and crash-triggered SOS alerts.

Finding	Website implication
Two-wheeler riders form a major share of road deaths	Position SafeRide as a rider-safety product, not a gadget.
Helmet non-use remains a major risk	Make strap-checking a core feature.
Smart helmets can be expensive	Emphasize an affordable retrofit module.
Prototype issues included false alarms, bulk, and battery limitations	Show testing honestly and improvement roadmap.

3. Goals and success metrics

Goals: Explain the product in under 10 seconds; communicate the strap-check and crash-SOS workflow; establish technical credibility; generate pilot sign-ups.

Metrics: CTA clicks, form submissions, demo-video engagement, mobile bounce rate below 55%, page load under 3 seconds, and form completion above 60%.

4. User stories

User	Need	Website outcome
Daily commuter	Understand usefulness and affordability	Clear benefits, price target, compatibility
Parent	Know whether family can be alerted	Emergency contacts, GPS sharing, privacy controls
College evaluator	Assess innovation and feasibility	Problem, design process, prototype, roadmap
Delivery rider	Understand daily-use practicality	Battery, durability, comfort, false-alert mitigation
Potential partner	Evaluate scalability	Architecture, roadmap, contact details

5. Website information architecture

Home → Problem → Solution → How It Works → Features → Prototype → Safety & Limitations → Roadmap → FAQ → Contact / Join Pilot

6. Page requirements

Home: Hero with “Ride Protected. Get Help Faster.”, primary CTA “Join Pilot”, problem snapshot, solution overview, how-it-works preview, benefits, prototype credibility, final CTA.

Problem: Explain helmet non-use, delayed help, and the limits of a conventional helmet. Use dated, sourced statistics.

Solution: Explain strap-check, crash detection, 10-second cancel window, SOS alert with location, and low-cost retrofit design.

How it works: Wear and buckle → Ride safely → Crash detected → Cancel or SOS sent.

Features: Strap-check alert, crash detection, cancel window, emergency-contact alerts, GPS sharing, buzzer feedback, sleep mode, companion app, ride history, fleet dashboard.

Prototype: ESP32, accelerometer, strap switch, GPS/GSM or BLE, buzzer, battery, casing; prototype photos, flowchart, testing results, and improvements.

Safety and limitations: State that SafeRide does not replace a BIS-certified helmet; crash detection may produce false positives/negatives; GPS and network connectivity can fail; location sharing requires consent.

7. Functional requirements

ID	Requirement	Priority
FR-1	Homepage explains product within first screen	P0
FR-2	Visual “How It Works” flow	P0
FR-3	Problem statistics with sources and dates	P0
FR-4	Prototype gallery	P0
FR-5	Pilot signup form: name, email, phone, rider type, city, message, consent	P0
FR-6	FAQ covering safety, battery, compatibility, privacy, limitations	P0
FR-7	Fully responsive design	P0
FR-8	Contact section with email and social links	P1
FR-9	Downloadable project report or pitch deck	P1
FR-10	Dark/light mode	P2
FR-11	Admin view for pilot sign-ups	P2

8. Non-functional requirements

Performance: First Contentful Paint under 1.8 seconds; page weight under 1.5 MB. Accessibility: WCAG 2.1 AA contrast, keyboard navigation, semantic HTML, alt text. Security and privacy: HTTPS, validated forms, spam protection, explicit location/contact consent. SEO: unique title, meta description, Open Graph tags, semantic headings, image alt text.

9. Homepage copy

Hero: Ride Protected. Get Help Faster.
SafeRide is a low-cost smart helmet add-on that checks your strap and can send an SOS alert with your location after a serious crash.

Problem: A helmet protects your head. It does not check whether you are wearing it properly—or call for help.

Solution: A small module. Two critical safety checks.

CTA: Join the Pilot.

10. VS Code website-generation prompt

Paste the prompt below into GitHub Copilot, Cursor, Codeium, or another AI coding assistant in VS Code.

Create a modern, clean, fully responsive marketing website for a college innovation project called “SafeRide” — a low-cost smart helmet add-on for motorcycle and scooter riders in India.

Tech stack:

- Use HTML5, CSS3, and vanilla JavaScript only.

- Create separate files: index.html, styles.css, script.js.

- Use CSS custom properties for a consistent design system.

- Do not use paid APIs, backend, database, or external frameworks.

- Make the site mobile-first and responsive from 360px to 1440px+.

- Add subtle scroll-reveal animations using Intersection Observer.

- Add a sticky responsive navbar with a mobile hamburger menu.

- Add smooth scrolling for anchor links.

- Use accessible semantic HTML, alt text, labels, visible focus states, and sufficient color contrast.

Brand direction:

- Product: SafeRide, a smart helmet safety module.

- Tone: trustworthy, modern, safety-focused, student-innovation friendly.

- Primary color: deep navy or charcoal (#0B1F33).

- Accent color: safety orange (#FF6B1A).

- Secondary colors: white, light gray (#F5F7FA), muted slate gray (#5B6B7A).

- Typography: “Poppins” for headings and “Inter” for body text from Google Fonts.

- Design style: generous white space, rounded cards, soft shadows, clean icons, professional startup landing-page look.

- Do not make it look childish or overly futuristic.

Website sections:

1. Sticky navbar: logo “SafeRide”; links: Problem, Solution, How It Works, Features, Prototype, FAQ, Contact; CTA “Join Pilot”; mobile hamburger menu.

2. Hero: badge “Smart Helmet Safety Add-On”; H1 “Ride Protected. Get Help Faster.”; subheading “SafeRide is a low-cost smart helmet add-on that checks your strap and can send an SOS alert with your location after a serious crash.”; CTAs “Join the Pilot” and “See How It Works”; clean CSS-based helmet/module visual; trust badges: Low-Cost Retrofit, Crash SOS, Strap Check.

3. Problem: heading “The Safety Gap”; explain that helmets protect the head but cannot ensure correct use or call for help after a crash; three animated statistic cards: 46.2% two-wheeler share of 2024 road deaths, 54,122 helmet non-use deaths in 2024, 10-second SOS cancel window; source note for MoRTH and NCRB.

4. Solution: heading “One Small Module. Two Critical Safety Checks.”; two large cards: Strap Check and Crash SOS; explain 10-second cancel countdown and GPS location alert; add note that SafeRide does not replace a certified helmet.

5. How It Works: 4-step timeline: Wear & Buckle, Ride Safely, Crash Detected, SOS Sent; add disclaimer that crash detection is an assistance feature and may not detect every crash.

6. Features: six cards: Strap-check alert, Crash detection, 10-second cancel window, Emergency-contact SOS, GPS location sharing, Low-power sleep mode; include compatibility note.

7. Prototype: two-column layout with prototype description and component list; components: ESP32, accelerometer, strap sensor, GPS + GSM or BLE, buzzer, rechargeable battery, compact casing; include testing and improvements: false alerts from speed breakers, higher threshold, cancel window, smaller board, sleep mode; target price Rs 1,000-Rs 1,500.

8. FAQ accordion: Does SafeRide replace my helmet? What happens after a crash is detected? Will it send false alerts? Does it work without mobile network? Is my location private? How much does it cost?

9. Contact / Join Pilot: form with full name, email, phone, rider type dropdown, city, message, consent checkbox; JavaScript validation; success message; temporarily store submissions in localStorage.

10. Footer: logo, description, quick links, contact email hello@saferide.example, social icons, copyright, and disclaimer.

Additional requirements:

- SEO title: “SafeRide | Smart Helmet Strap Check & Crash SOS Add-On”

- Meta description: “SafeRide is a low-cost smart helmet add-on that checks helmet straps and sends SOS alerts after a detected crash.”

- Add Open Graph tags and inline SVG favicon.

- Add scroll progress bar and back-to-top button.

- Add dark/light mode toggle.

- Ensure hover and focus states for all interactive elements.

- Avoid placeholder text such as “Lorem ipsum.”

- Make the site polished enough for a college project presentation, hackathon, or startup pitch.